

Lecture 4

Metric Space

- A set M with a metric $d : M \times M \rightarrow \mathbb{R}$
- $\forall x, y, z \in M$, 3 properties hold:
 - Positive Definite: $d(x, y) \geq 0$ & $d(x, y) = 0$ iff $x = y$
 - Symmetric: $d(x, y) = d(y, x)$
 - Triangle Inequality: $d(x, z) \leq d(x, y) + d(y, z)$
- Denoted as (M, d)
- Note on $\emptyset$: Generally not thought of as metric space, but it can fulfill conditions vacuously with empty function as metric
- Isomorphism: Existence of metric in one set means isomorphic space also has metric

Examples

1. $\mathbb{R}, d(x, y) = |x - y|$
2. $\mathbb{R}^m$, Euclidean distance
3. If M is a metric space & $S \subset M$, d_S on $S \times S$ is $d_{S(x,y)}$ where $x, y \in S$ and d_S is the inherited metric from M
 - $[a, b] \subset \mathbb{R}$
 - A disc in the plane
 - d_D inherited from plane, same as Euclidean Distance
4. Discrete Metric on S
 - $$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$
 - Think of dots, “distance” to all other points is 1
5. Cartesian product of 2 metric spaces
 - (M_1, d_1) and (M_2, d_2)
 - 2 separate metrics $d_1(x_1, y_1)$ and $d_2(x_2, y_2)$, how to combine?
 - $d_1(x_1, y_1) + d_2(x_2, y_2)$
 - $d_{\mathbb{R}}(d_1(x_1, y_1), d_2(x_2, y_2))$
 - $d_{\mathbb{R}^2}((0, 0), (d_1(x_1, y_1), d_2(x_2, y_2)))$
6. S^2 , metric would be great circle
7. T^2 (torus), metric is shortest curve connecting 2 points
 - Similar to line connect points on annulus

Sequences

- (a_n) is a sequence in a set S
 - Sequence refers to infinite sequence, finite sequence will be specified
- Comparison
 - $(2, 4, 6, 8, \dots)$ is a sequence in $\mathbb{N}$
 - $\{2, 4, 6, 8, \dots\}$ is a set in $\mathbb{N}$, which lacks an ordering
 - Ordering also allows for sequences to have repetition
- Can be formally defined as a $f : \mathbb{N} \rightarrow S$
- Series are infinite sums
 - $\sum_{k=1}^{\infty} a_k$

Convergence

- Sequence (a_n) on M converges to a point $a \in M$ iff $\forall r > 0, \exists N \in \mathbb{N}$ s.t. $\forall n \geq N, d(a_n, a) < r$

- r can be replaced with ε here
- r -neighborhood of a , denoted as $M_{r(a)}$, is defined to be $\{x \in M : d(x, a) < r\}$
- For limits, we want a_n to be in the r -neighborhood of a for some large n
 - Shows why limit is unique
 - Suppose there are 2 different limits. Then, there is some distance between the two limits. Once r is less than half the distance between the 2 limits, the r -neighborhoods are disjoint, and a point in the sequence can't be in both.

Theorem 1: Every bounded monotonic sequence in $\mathbb{R}$ converges to a limit in $\mathbb{R}$

Proof: It's obvious

- For a monotonically increasing sequence, $\lim_{n \rightarrow \infty} (a_n) = \text{LUB}(\{a_n\})$

Theorem 2: Every sequence in $\mathbb{R}$ has a monotone subsequence

Proof: Peaks

- Subsequences vary on a sequence on n
 - Sequence is $(a_n)_{n=1}^{\infty}$, subsequence is $(a_{n_k})_{k=1}^{\infty}$ where $n_1 < n_2 < n_3 < \dots$
- a_n is a peak if $\forall m \in \mathbb{N} (m > n \rightarrow a_m \leq a_n)$
- A diagram would be nice here but oh well
- 2 cases
 - Infinitely many peaks
 - Implies that we can start with any peak and choose the peaks that come after to create a monotonically decreasing subsequence
 - Finitely many peaks
 - Look after last peak
 - Implies that a monotonically increasing sequence exists following the location of the peak, as no peaks exist, so there must always be some value a_m greater than the current value but less than a_n

Corollary 1: Every bounded sequence in $\mathbb{R}$ has a convergent subsequence

- Apply idea of peaks to first theorem
- Bounded sequence has a monotone subsequence (by theorem 2), so this bounded monotone subsequence converges

Compactness

- M is compact iff every sequence (a_n) in M has a subsequence (a_{n_k}) that converges to a point in M
- Example: $[a, b]$
- Nonexample: (a, b)

Cantor's Theorem Proof

- $|S| < |\mathcal{P}(S)|$
- Equivalent to saying $\nexists$ bijection on $S \rightarrow \mathcal{P}(S)$
 - Injection from S to $\mathcal{P}(S)$, can map each element $s \in S$ to $\{s\} \in \mathcal{P}(S)$
 - No bijection means no injection from $\mathcal{P}(S) \rightarrow S$
- AFSOC, β is a bijection from $S \rightarrow \mathcal{P}(S)$
- Split S into two parts, B and B^C
- Suppose $\forall x \in B (x \notin \beta(x))$
- $B \subset S$, so $B \in \mathcal{P}(S)$
 - $\exists \xi \in S (\beta(\xi) = B)$

- $\xi \in B^C$
- Something something QED!